

Aether Dark

Dual-voice drone instrument · User manual

Aether Dark is a browser instrument: two analog-style voices, a shared Drift LFO, noise, a wavefolder, tape print, echo, bloom, a patchbay, and Evolve — a slow climate layer so the sound can change over minutes without becoming a different patch. Manual for build 2026-09-25k.

Play it as a webpage (GitHub Pages <https> link) or as a downloaded index.html in desktop Chrome/Edge. Tap POWER once. MIDI (including the Akai MIDI Mix map) works in those desktop browsers. iPad: open the <https> link in Safari and keep the tab in front — do not open the file from the Files app preview. Web MIDI is not available on iOS. Support: ko-fi.com/tribalvoice

Not affiliated with Dubreq or Stylophone. Original Aether Series instrument.

1. Signal flow

Voice I and Voice II each mix square, triangle, sine and saw with a triangle sub. Fold, if above zero, reflects that mix before a soft shaper and a biquad (LP/HP). Both voices hit the mix bus → echo → bloom → optional tape print → rumble high-pass → main out. Noise can join the mix or its own copy of Filter I / Filter II. Drift is a single-rate LFO with two waveforms. Evolve does not add an oscillator; it walks offsets around the panel values. A teal tick on a knob is the sounding lean; the silver pointer stays on the saved patch.

2. Drift LFO

One rate, two shapes. Destination toggles choose whether each voice's osc and filter listen to Wave I or Wave II.

Control	What it does
Wave I / Wave II	Stepped shapes including Off, triangle, square, saw, multiples. Click the knob to cycle.
Rate	0.02–70 Hz. The swell / pulse of the instrument.
OSC / FILTER dest	Up = Wave II, down = Wave I.
RATE jack	CV in, scales LFO rate.
RESET jack	Rising edge restarts both phases.
WAVE I / II jacks	CV out of the current LFO sample.

3. Evolve — climate, not a second LFO

Drift is the wave you already hear. Evolve is weather over a minute or several minutes so the next swell is not a photocopy of the last. Depth 0 leaves every factory patch unchanged.

3.1 Controls

Control	Range	Role
Time	45 s – 8 min (log)	How long the climate takes to lean to a new target, then pick another.
Depth	0–1, default 0	How far Evolve may stray from the panel knobs. The silver pointer stays put. A teal tick shows the sounding lean.
Target	Tone / Weather / Motion / All	What it is allowed to touch.
Irregular	0–1, default 0	On each wrap of Wave I, dice a new offset for filter LFO depths and noise LFO depth. Separate from Time.
Loop / Once	toggle	Loop (down) keeps wandering. Once (up) takes one journey and holds.

CV jack	the walk, about $-0.9 \dots +0.9$	See 3.6. This is the climate itself, not audio.
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3.2 Targets

Target	Moves
Tone	Both X-Waves, both Folds, both cutoffs. Never pitch or volume.
Weather	Noise level only. Colour, dest and tape stay where you set them.
Motion	Drift Rate and Noise LFO Rate $\times$. The pulse speeds up or dies.
All	Tone + Weather. Rate is left alone so the pulse stays your pulse.

3.3 How it moves

Not a triangle. A slow random walk: Evolve eases toward a hidden aim between about -0.9 and $+0.9$, then chooses a new aim. Time is how long that journey takes. Depth scales how far the lean may go. A short Time is a faster, more obvious change; a long Time is climate. Each Loop lap rolls a new aim, so two passes are cousins, not copies. Once takes one journey and holds. Turning a knob still writes the saved patch — the tick shows where the sound has leaned.

Irregular is the “each wave is a different wave” control. 0 = identical cycles. Noon = set-to-set change of swell height. Full = almost sample-and-hold on how far this crash opens.

3.4 Listening examples

Shoreline + Time ~ 2 min, Depth $\sim 30\%$, Target Weather — shore stays shore, hiss fattens, print ages.

Breakers + Irregular at noon — crashes not the same height.

Chapel Fifth + Target Tone, Depth low, Once — the fifth slowly blooms and stays open.

Factory patch Long Climate is Tide/shore language with All + Irregular already on. Deep Tide also has a gentle All wander. Leave them running; do not judge in ten seconds.

3.5 What Evolve does not do

No second waveform bank. No tempo sync. No morphing one factory patch into another. No Main Out automation. Motion is the only target that changes the clock.

3.6 Evolve CV output

The jack under Evolve is control voltage, not audio. It sends the walk itself (about -0.9 to $+0.9$), whether or not Depth is up. Depth only scales how far the panel leans. The jack does not scale with Depth, so a cable can still move something when the ticks are almost still.

Patch it like any other CV out: click the Evolve CV jack, then a CV in. Useful destinations:

Into	What you hear
Filter CUTOFF	About ± 2 octaves of extra cutoff travel, on top of the LFO and the internal Tone lean.
PITCH	About ± 0.75 octave. Slow, not a melody. Two linked voices move together.
PWM	X-Wave / pulse travel if that input is in use.
LFO RATE	The swell speeds up or dies with the climate. This is Motion, done with a cable.
LFO RESET	A rising edge (walk crossing about $+0.2$) restarts both LFO phases. Rare, and easy to overdo.

The walk is not seeded. You cannot reprint the same cable performance by reloading the patch. For a fixed bed, leave the cable out and print once.

4. Voices

Each voice: Osc Mix (X-Wave ↔ sub), Sub -1/-2, X-Wave, Fold, LFO → pitch or X-Wave, Pitch + tuner, Filter resonance / LP-HP / LFO depth, Cutoff, Volume. Voice II adds Hard Sync, Pitch Link, and Offset in semitones. Jacks: PITCH, PWM, CUTOFF, OUT.

Fold sits under X-Wave. At 0 the dry path is unchanged. Turned up, the mix is reflected back on itself before the soft clip and the filter — extra harmonics, more metal, then the filter can darken it again. Dusk sine with a little Fold is the soft way in. Full Fold into a resonant low-pass is the raspy way. Evolve Tone and All lean on Fold; the teal tick shows it. Old presets load at 0.

Dark / Dusk is the switch beside Rumble and FX/Dry, and it is saved with the patch. Dark is the three-way blend (square–triangle–saw). Dusk inserts sine in the middle of the travel (square–triangle–sine–saw). The sine oscillator is already in the voice; Dark simply turns its gain to zero. Factory patches load as Dark unless you saved them otherwise.

A long pitch sweep with Echo up sounds like a spring tank being knocked: the delay still holds the old pitches. Pitch Link makes that worse because the fifth moves together. Dry the echo if you want a clean glide. Hard Sync forces Voice II to whole-number harmonics of Voice I, which is why a linked sweep can sound stepped.

5. Noise

Control	What you hear
Level	How much noise. Independent of voice volume when Dest is a filter.
Colour	White / Pink / Brown = textures. S&H = stepped chunks of hiss (a new loudness each clock), not a DC voltage.
Dest	Off / Mix / Filter I / Filter II. Filter dest uses a copy of that voice's cutoff, Q and LP/HP.
LFO Depth	Swell of noise level from Wave I. 0 = flat.
LFO Rate ×	0.25×–4× Drift Rate.
Phase Offset	0–360°. 180° puts the crash in the trough of the voice swell.
OUT	Audio of the noise generator. Also the CV source described in 5.1.

S&H clock is Drift Rate × Noise LFO Rate. Embers uses it. Other colours are continuous textures, not steps.

5.1 Noise as a CV output

The Noise jack is labelled OUT and is an audio output: the coloured noise, after level, heading for the mix or a voice filter depending on Dest. You can also click it as a patch-cable source. In that role it does not send the hiss. It sends the sample-and-hold value — a bipolar step, about -1 to +1 — and only while Colour is S&H.

If Colour is White, Pink or Brown, that CV sits at 0 (or on the last step you held). Switch to S&H before you expect the cable to move. The step clock is Drift Rate × Noise LFO Rate ×. Phase Offset does not move the hold clock; it moves the amplitude LFO of the hiss.

Same destinations as Evolve CV, different gesture: Evolve is a slow walk; Noise S&H is a new random voltage each Drift cycle (or faster, if Rate × is above 1). Into cutoff it is a stepped filter. Into pitch it is a stepped interval, about ±0.75 octave. Into LFO rate it makes the swell stutter. It will not do wind or rain — that is the audio OUT into Mix or a filter, not the cable.

6. Mixer, echo, bloom, tape

Delay D/W, Time, Repeats, echo LFO depth and Wave I/II source. Bloom is the reverb: Algorithm picks the room, Reverb D/W is how wet it is. Dry voices sit in the centre; Bloom's wet tail is stereo. Tape: Print (level-matched sat — Print off mutes the wet path, it does not unload the tape graph), Sat, Wow, Wear. Wow is subtle at low settings and still moves noise more than the oscillators; keep it low on drone presets.

Under the tape block: Rumble HPF (high-pass on the main out, default 22 Hz — turn it up to thin the subs), FX/Dry (bypass echo and bloom, tape still in the path if Print is on), and Dark/Dusk. The meter follows the main out; full panel levels do not pin it.

REC captures the printed mix. SAVE writes a 16-bit mono WAV. PLAY auditions the last take; with POWER off it plays the take only and does not start the drones. For a production stereo bounce, record the browser output in a DAW (Audacity on Windows: WASAPI loopback), not SAVE.

7. Patchbay

Click an output, click an input. Click a cable to remove it. Clear Cables in the toolbar. Audio outs: Voice I OUT, Voice II OUT, Noise OUT (the hiss). CV outs: Wave I, Wave II, Evolve CV (the walk), and Noise OUT when Colour is S&H (the hold, not the hiss). CV ins: LFO RATE, LFO RESET, voice PITCH, PWM, CUTOFF.

8. MIDI

Generic map: CC1 rate, CC7/11 level, 71/74 Voice I res/cut, 76/77 Voice II, 91 bloom, 94/72/73 echo. Notes tune the drones. Program change loads factory patches in list order.

Map: Akai MIDI Mix uses the Mix factory CCs. Mute/Rec notes are buttons, not pitches. Do not hit Send All unless you want every parameter slammed. Bank left/right step factory patches if they send notes 25/26.

9. Factory patches (selected)

Patch	Idea
Ambient Wash / Stereo Hall / Chapel Fifth	Slow linked drones, little or no noise.
North Wind / Gale / Shoreline / Breakers	Noise as weather. Breakers pulses brown through Filter I, 180° offset.
Rain / Thunder Bank / Night Insects / Embers	Detail weather. Embers = S&H into Filter II.
Deep Tide / Long Climate / Fog / Glacier	Leave running. Climate and Tide wander under Evolve.

10. Hosting and files

One GitHub repository holds everything. Name the instrument index.html so Pages can serve it. Enable Settings → Pages → deploy from the main branch, folder / (root). The play URL is <https://YOURUSER.github.io/REPO/>.

iPad and any browser: that https link. PC offline: download index.html from the repo (or a Release) and open it in Chrome. Same file, two ways to reach it. Put this PDF in the repo root as Aether-Dark-User-Manual.pdf so the Manual link on the page resolves.

Do not use the iPad Files preview. If knobs are dead, close that preview and open the Pages URL in Safari. The orange banner can be dismissed; it also hides after POWER succeeds, and on http/https.

11. Support and patches

The instrument is free. Ko-fi: ko-fi.com/tribalvoice (toolbar chip and footer). Randomise includes noise, Evolve and sometimes Dusk. Patches live in this browser's localStorage; Export JSON to move them between machines.